

that contain four fused carbon rings; all are derived from the steroid cholesterol (see Figure 5.12). Epinephrine and thyroxine are amine hormones, each synthesized from a single amino acid, either tyrosine or tryptophan.

As Figure 45.4 indicates, hormones vary in their solubility in aqueous and lipid-rich environments. Polypeptides and most amine hormones are water-soluble, whereas steroid hormones and other largely nonpolar (hydrophobic) hormones, such as thyroxine, are lipid-soluble.

Cellular Hormone Response Pathways

Water-soluble and lipid-soluble hormones differ in their response pathways. One key difference is the location of the receptor proteins in target cells. Water-soluble hormones are secreted by exocytosis and travel freely in the bloodstream. Being insoluble in lipids, they cannot diffuse through the plasma membranes of target cells. Instead, these hormones bind to cell-surface receptors, inducing changes in cytoplasmic molecules and sometimes altering gene transcription (Figure 45.5a). In contrast, lipid-soluble hormones exit endocrine cells by diffusing out across the membranes. They then bind to transport proteins, which keep them soluble in blood. After circulating in the blood, they diffuse into target cells and typically bind to receptors in the cytoplasm or nucleus (Figure 45.5b). The hormone-bound receptor then triggers changes in gene transcription.

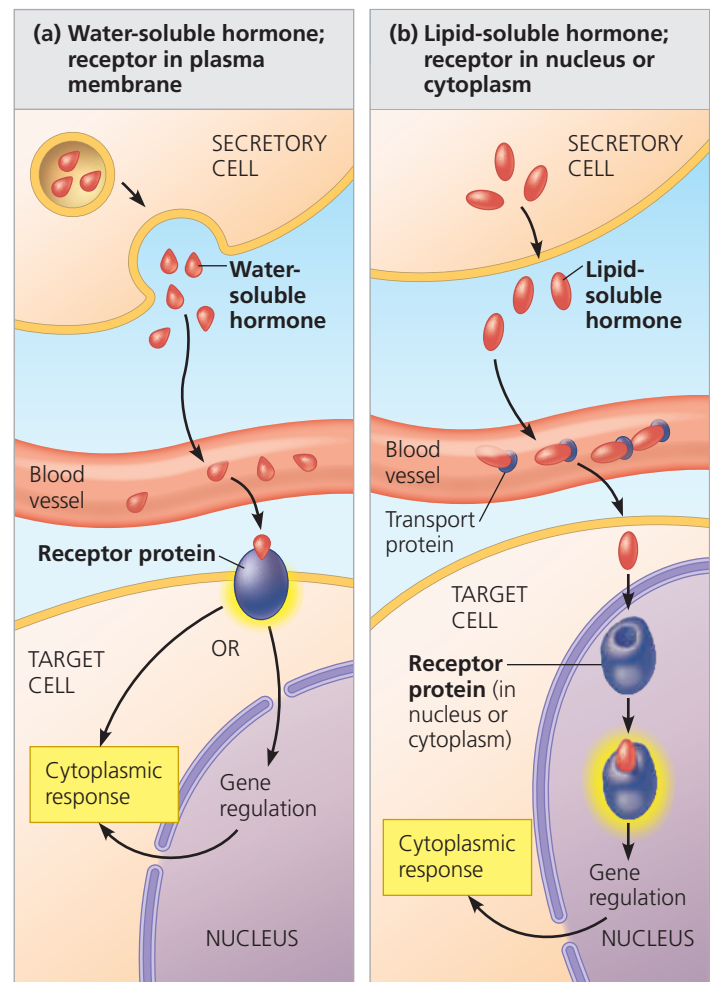
To explore further the distinct cellular responses to water-soluble and lipid-soluble hormones, we'll examine the two response pathways in turn.

Response Pathway for Water-Soluble Hormones

The binding of a water-soluble hormone to a cell-surface receptor protein triggers a cellular response. The response may be the activation of an enzyme, a change in the uptake or secretion of specific molecules, or a rearrangement of the cytoskeleton. In some cases, cell-surface receptors cause proteins in the cytoplasm to move into the nucleus and alter the transcription of specific genes.

The chain of events that converts the extracellular chemical signal to a specific intracellular response is called **signal transduction**. As an example, we'll consider one response to short-term stress. When you are in a stressful situation, perhaps running to catch a bus, the adrenal glands that lie atop your kidneys secrete the water-soluble hormone **epinephrine**, also known as *adrenaline*. Epinephrine regulates many organs, including the liver, where it binds to a G protein-coupled receptor in the plasma membrane of target cells. As shown in Figure 45.6, this interaction triggers a cascade of events involving synthesis of cyclic AMP (cAMP) as a short-lived *second messenger*. Activation of protein kinase A by cAMP leads to activation of an enzyme required for breakdown of

▼ **Figure 45.5** Variation in hormone receptor location.



WHAT IF? Suppose you are studying a cell's response to a particular hormone. You observe that the cell produces the same response to the hormone whether or not the cell is treated with a chemical that blocks transcription. What can you surmise about the hormone and its receptor?

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glycogen into glucose, as well as inactivation of an enzyme needed for glycogen synthesis.

Note that there are three enzymes in this signal transduction cascade—adenylyl cyclase (which converts AMP to its cyclic form), protein kinase A, and, for example, the enzyme that breaks down glycogen into glucose. Each enzyme-catalyzed step in the cascade provides an opportunity for signal amplification: One enzyme molecule can catalyze many reactions, thereby generating multiple signals at that step in the cascade. Furthermore, because the three enzymes act at different steps in the same pathway, the net effect can be enormous. If, for instance, each enzyme carried out 1,000 reactions, the binding of one molecule of epinephrine to its receptor would trigger cleavage of a billion ($10^3 \times 10^3 \times 10^3$) glycogen molecules. The net result is that the liver releases a substantial quantity of glucose into the bloodstream, quickly providing the body with extra fuel.